Predicting User Trust in Customer-Service Chatbots: A Supervised Learning Study

Abstract

User trust is the basis of chatbots for customer-service, especially where disclosure of personal information is in question. We frame trust prediction as supervised binary classification on questionnaire data (N=122) with questionnaire items (demographics, frequency of use, overall and recent satisfaction, and sector exposure). The binary label is the will to disclose personal information (Yes/No). We experiment with Logistic Regression, Random Forest, and HistGradientBoosting with stratified 5-fold cross-validation and a 20% held-out test set and report Accuracy, F1, Precision, Recall, ROC–AUC, and PR–AUC. HistGradient-Boosting performs best on the test set (Accuracy 0.88, F1 0.857). Feature attribution across models reveals overall and recent satisfaction, frequent use, and sector context (e.g., bank/healthcare) as top indicators. We calibrate predicted probabilities and propose a conservative escalation rule—handoff to a human where P(trust) < 0.35—to keep low errors in high-risk sectors very sensitive to privacy. Calibrated predicted probabilities and the rule we propose are our contributions towards a safer, sector-aware chatbot design and a set of actionable recommendations for deployment in practice.

Keywords

Chatbots; Trust Prediction; Supervised Learning; Customer Service; Probability Calibration; Human-in-the-Loop

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